UNIT 1

COMPUER SYSTEM

1.0 Introduction to the computer system.

Computer (The word 'Computer' comes from the Latin word 'Computare' meaning 'to calculate.')

Definition: A computer is an electronic device that accepts raw data, processes it according to a given set of instructions, stores it for future use, and provides meaningful information.

Features of a computer /characteristics of a computer

(i) Automatic: Performs tasks automatically without further human involvement.

(ii) Speed: works at high speed, performs millions of calculations in a few seconds.

(iii) Accuracy:100% accurate device, produces incorrect output due to incorrect input (GIG0)

(iv) Storage: Can store large amounts of data in storage devices

(v) Diligence: Works repeatedly with same speed or accuracy for a long time.

(vi) Versatile: Can be used for many different types of tasks.

Functions of a computer system: Input, Process, Output, Storage (IPOS) with Basic Block Diagram

{(Q) Explain the working principle of a computer system with the help of a block diagram and describe it in details.}

A computer system works on a systematic process called the IPOS cycle, which stands for Input, Process, Output, and Storage. The block diagram of IPOS cycle of computer system be given below.

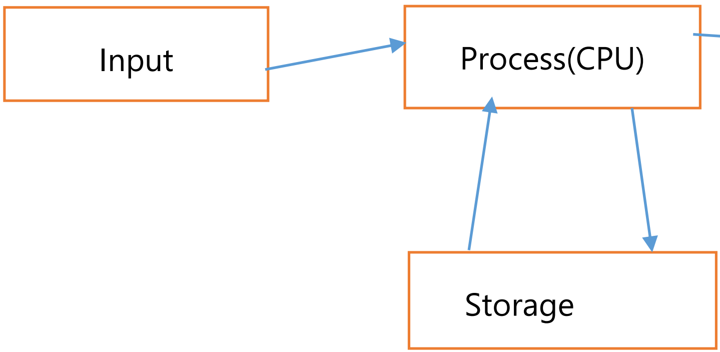


Fig: Block Diagram of the IPOS cycle.

Input: The computer accepts raw data and instructions from the user through various input devices such as a keyboard, mouse, scanner, etc. In simple words, input is the process of giving data to the computer.

Process: The CPU processes the raw data according to the given set of instructions and converts it into meaningful information. In simple words, a process means converting raw data into useful information.

Output: The computer displays the meaningful information to the user through output devices like a monitor, printer, or speaker. In simple words, output is the final result shown by the computer.

Storage: The computer stores data and information for future use, so it can be accessed later when needed.

Application Areas of the Computer System

Due to the versatile capabilities of computers, they are widely used to perform different kinds of tasks. The main application areas of computers are described below.

1. Education: Computers are widely used in education for teaching and learning. Students can study using the internet, videos, and online classes.
2. Communication: Computers help people send messages, emails, and make video calls through the internet quickly and easily.
3. Banking: Computers are used in banks to manage accounts, transfer money, and provide online banking services safely.
4. Medicine: Doctors and hospitals use computers to check patients' health and treat diseases. Computers also help in tests such as X-rays and CT scans.

5. Entertainment: Computers are used for entertainment purposes such as playing games, watching movies, listening to music, and using social media.
6. Research: Scientists use computers for research and experiments. Computers help to study and analyze large amounts of data easily.
7. Agriculture: Computers help farmers in modern farming by checking the weather, controlling irrigation, and managing crops.

(Q) Describe the use of computers in the education field.

Computers play a very important role in the field of education. Some common use of computers in the education field are given below.

1. Teaching and Learning: Computers are used in classrooms to teach students through videos, animations, and presentations. This makes learning more clear and easy to understand.
2. Online learning: Students can learn from home using computers and the internet. They can join online classes, watch tutorials, and study different subjects anytime.

3. Preparing study materials: Teachers use computers to prepare notes, question papers, assignments, and presentations.
4. Examination and results: Computers are used to prepare question papers, conduct examinations, calculate marks, and publish examination results accurately and quickly.

Short Que /Ans (imp for exam)

Q1. Define the computer.

Ans: A computer is an electronic device that accepts raw data, processes it according to a given set of instructions, stores it for future use, and provides meaningful information.

Q2. What is IPOS? How do IPOS work?

Ans: IPOS stands for Input, Process, Output, and Storage. The computer accepts raw data (Input), the CPU processes it (Process), shows the result (Output), and saves it for future use (Storage).

Q3.What is GIGO?

Ans: Due to incorrect input, a computer may produce incorrect output. This is called GIGO (Garbage In, Garbage Out)

Q4. List any four computer features.

Ans:

Automatic — performs tasks automatically - - -

Speed — works at very high speed (nanoseconds)

Accuracy — 100% accurate

Diligence — works repeatedly without losing speed or accuracy

Q5.Why is a computer called a diligent machine?

Ans: A computer is called a diligent machine because it can perform tasks repeatedly for a long time with same speed and accuracy. Unlike humans, it does not get tired or bored. This ability is known as diligence.

Q6. List four computer related uses.

Ans: Education, Communication, Banking, Agriculture

Q7. Identify distinct computer storage units?

Ans: Bit , Byte ,KB ,MB ,GB ,TB , Peta Byte , Exa Byte , Zetta Byte, Yotta Byte.

Long Que/Ans (Imp for exam)

Q1. Explain the working principle of a computer system with the help of a block diagram and describe it in details.

Q2. Describe the use of computers in the education field.

1.2 Input device

Definition: Input devices are tools used to enter raw data and instructions into a computer system.

For example: Keyboard, mouse, microphone, joystick, scanner etc.

Some examples:

1. Mouse

A mouse is a handheld pointing device used to interact with a computer. A typical mouse has three buttons:

Left button

Right button

Scroll button

2. Keyboard

A keyboard is an input device used to type letters, numbers, and commands into a computer.

A standard keyboard consists of 104 keys.

Types of Keys:

Alphabet keys – A to Z and punctuation symbols

Numeric keys – 0 to 9 and mathematical symbols

Function keys – F1 to F12

Cursor movement keys – Arrow keys (← ↑ → ↓)

Special purpose keys – Enter, Spacebar, Shift, Ctrl, Alt

3. Joystick

A joystick is an input device mainly used for playing video games.

It has a movable stick and buttons for controlling actions in games.

4. Microphone

A microphone is a input device used to capture sound and convert it into digital form.

5.Scanner

A scanner is a input device used to convert physical documents or pictures into digital format and store them in a computer

6 . webcam

A webcam captures real-time video and audio. It is used for video calls, online classes.

7. Touchscreen

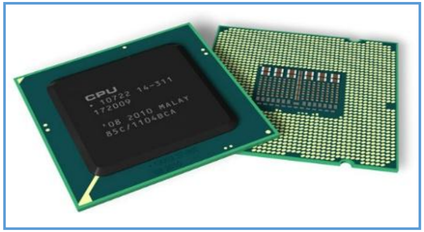
A touchscreen allows users to give commands by touching the screen.

8. Touchpad

A touchpad is found in laptops which acts like a mouse.

1.3 Central Processing Unit CPU

Definition: CPU is known as the 'brain' of the computer. It receives data and instruction from input devices, Executes instructions, and generates output.



Functions of the CPU

1. Fetching: The CPU collects data and instructions from memory (RAM). (collect 2, 3, add or +)
2. Decoding: The CPU interprets the fetched instructions to understand what Actions are required. (CPU interprets + as addition)
3. Executing: The CPU performs the necessary operations such as calculations, data processing, or interaction with other hardware. (performs 2+3)
4. Storing: After processing, the CPU stores the result in memory or sends it to an output device.(store 5 and display on monitor)
5. Managing interrupts: The CPU handles hardware and software interrupts to ensure smooth, continuous processing.

Components of CPU/ Architecture of CPU

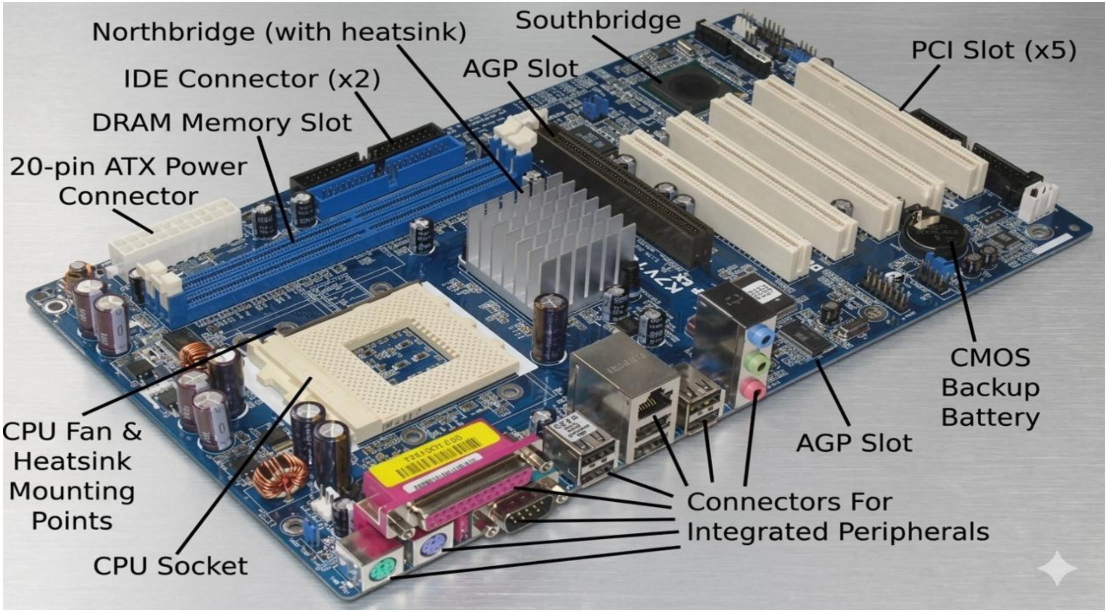
The CPU is known as the “brain” of the computer. It has three main components that work together to process data.

1. Control Unit (CU): The Control Unit controls the working of all parts of the computer system. It manages the flow of data inside and outside the CPU. It communicates between the registers and ALU, and also between all input, output, and storage devices.
2. Arithmetic Logic Unit (ALU): The ALU performs all arithmetic operations such as addition, subtraction, multiplication, and division. It also performs logical operations like greater than, less than, equal to, and not equal to.
3. Registers/Memory unit (MU): Registers are small memory units used to store immediate data, instructions.

1.4 Motherboards and Data Bus

Definition: The motherboard is also known as the PCB (Printed Circuit Board) or main circuit board.

It is the "backbone" of the computer because motherboard connects all components of a computer system and allows them to communicate with each other.



Functions of the Motherboard (serves as a central connector hub)

1. Physical connection: The motherboard connects components like the CPU, RAM, graphics card, keyboard, and mouse.
2. Electrical connections: Distributes power from the PSU; transfers data through circuits.
3. Communication Hub: Chipsets (Northbridge, Southbridge) manage data flow between CPU, RAM, and peripheral devices, such as the mouse and keyboard.

Data Bus

Definition: A Data Bus is an electronic pathway that transfers digital data between CPU, memory, and other components. It is made of parallel lines. The width of data bus (in bits) determines how much data is transferred at one time.

- Wider bus: = faster communication and better performance
- 3 types of buses: Data Bus, Address Bus, Control Bus

Function of Data Bus

1. Transferring Data

2. Handle Different Data Types

1.5 Computer memory

Memory stores data and instructions for processing. There are two types of memory.

1. Primary memory: Main memory; directly communicates with CPU; stores data temporarily; faster but expensive (RAM, ROM)
2. Secondary memory: Storage memory; stores data permanently; slower but cheaper (Pen drive, CD)

A. Primary Memory

Primary memory is the main/internal memory. It temporarily stores data and instructions while processing, and can directly communicate with CPU. Two types: (RAM, ROM)

RAM – Random Access Memory

Definition: RAM is volatile memory, all data stored in it is erased when the computer power is turned off. RAM can access data from any location randomly. It holds data and instruction during processing. Can read and write data.

Types of RAM:

SRAM (Static RAM): Type of RAM that stores data using transistors, fast, expensive, no need for periodic refreshing

DRAM (Dynamic RAM): Type of RAM that stores data using capacitors, slower, cheaper and needs periodic refreshing.

Difference: SRAM vs DRAM

SRAM	DRAM
It is made up of transistors.	It is made up of capacitors.
It is more expensive.	It is less expensive.
SRAM does not need periodic refreshment.	DRAM needs periodic refreshment.
SRAM is less dense.	DRAM is denser.
SRAM is faster than DRAM.	DRAM is slower than SRAM.
Data is stored in the form of voltage.	Data stored in the form of charge
Charge does not leak	Data charge gets leaked

ROM- Read Only Memory

Definition: ROM is a type of non-volatile memory, so the stored data remains safe even when the power is turned off. (This type of memory stores BIOS information needed to start the computer). Can read data only.

Types of ROM: PROM - Programmable ROM

EPROM - Erasable Programmable ROM

EEPROM – Electrically Erasable Programmable ROM

Difference between PROM, EPROM, EEPROM

PROM	EPROM	EEPROM
It can be programmed only once and cannot be erased after programming.	It can be erased using UV (Ultraviolet) light rays and can be reprogrammed many times.	It can be erased and rewritten using electrical signals. It is easy and fast to change.
Used in electronic devices, such as digital watches and calculators.	Mainly used in old computers.	Used in modern computers and smartphones.

Cache Memory

Definition: Cache memory is a high-speed memory located between the CPU and RAM. It stores frequently used data and instructions so that the CPU can access and process them quickly.

CPU working order: CPU first searches in Cache → if not found, searches in RAM → loads to Cache→ CPU access and process them quickly

Main purpose: To make CPU execution Fast.

Firmware

Definition: Firmware is a type of software permanently stored inside hardware devices, used to control and operate the hardware. For Eg. The BIOS program stored in ROM is firmware that helps to start the computer.

Primary vs Secondary Memory

Primary Memory	Secondary Memory
Also called main/internal memory	Also called auxiliary/backup memory
Communicates directly with CPU	Communicates indirectly with CPU
Temporarily stores data and instructions during processing.	Permanently stores data and instructions
It is faster than secondary memory.	It is slower than Primary memory.
More expensive	Less expensive
Cannot be physically transferred between computers(RAM, ROM)	Can be transferred (pen drive, CD, etc.)
Can be volatile (RAM) or non-volatile (ROM)	Always non-volatile
Less storage capacity	Very high storage capacity
Examples: RAM, ROM, Cache Memory	Examples: HDD, SSD, CD, DVD, Pen Drive, Cloud storage

1.6 Storage devices (Secondary memory)

Definition: Storage devices (secondary storage) store data and instructions permanently. So the stored data remains safe even when the power is turned off. For example CD, DVD

Types of Storage Devices

1. Hard Disk Drive (HDD)

A hard disk is a storage device used to store large amount of data permanently.

- It is made of aluminum with ferromagnetic coating,
- Inside, circular metal platters (disks) spin while Reading/writing (more power consumption)
- Speed: 5400 to 7200 RPM (Revolutions Per Minute)
- Capacity: 500 GB to 18 TB

2. SSD (Solid State Drive)

- Faster than HDD; no moving parts
- Uses semiconductor chips (NAND flash)
- More reliable, less power consumption
- More expensive than HDD
- Read/write speeds: 200 MB/s to 3500 MB/s
- Used in laptops, notebooks, Ultrabooks.

3. Optical Storage Discs

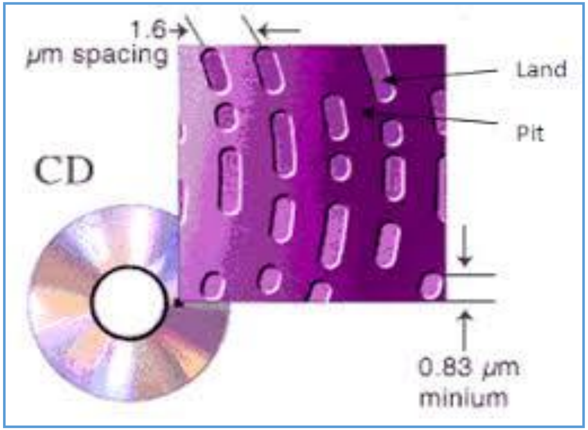
An optical storage disc is a disc that uses laser light technology to store and access data.

Types of storage Discs:

i. CD – Compact Disc: single-layer optical storage disc that uses laser light technology to store and access data.

ii. DVD -Digital Versatile Disc: ....

CD	DVD
Its storage capacity is less 700MB. (CD stores data in one single layer only.)	Its storage capacity is high up to 20GB. (DVD can store data in multiple layers.)
It is used to store audio and software.	It is used to store video and software.
It is cheaper.	It is expensive.
Developer: Philips and Sony (1980)	Philips, Sony, Toshiba, Panasonic (1995)
Types CD-R, CD-RW	Types DVD-RW, DVD+RW



Made of polycarbonate and coated

With Aluminium reflective laye**r.**

The various DVD storage capacities are:

- Single Side Single Layer – 4.7 GB
- Single layer Double Layer – 8.5 GB
- Double Side Single Layer – 9.4 GB
- Double Side Double Layer – 17.08 GB

4. Flash Memory and Pen drive

It is a non-volatile memory that can be electrically erased and reprogrammed many times. It stores data permanently, even when power is turned off.

Examples:

Pen drive, Memory card**,** Digital camera storage, SSD, Smartphone storage

Importance

- Stores data permanently
- Can be erased and rewritten many times
- *Portable and easy to use*

Cloud Storage

Cloud storage is the online storage of data on remote servers that can be accessed through the Internet.

Examples: Google Drive, Dropbox, Microsoft One Drive, Amazon S3

Importance:

• Can access data from anywhere

• Saves storage space on devices

• Cost-effective and flexible storage solution.

Note:

NAND Flash (NAND = special chip design Flash = quickly erasable memory)

It is a semiconductor memory chip that stores data permanently and can be erased/rewritten electrically.

HDD VS SSD VS Flash Memory Comparison

FEATURES	HDD	SSD	FLASH MEMORY
Technology	Spinning disks+ read/write heads	NAND flash, no moving parts	NAND flash, no moving parts
Speed	80–160 MB/s	200–3500+ MB/s	10–600 MB/s
Capacity	500 GB – 18 TB	Lower but increasing	Lower (portable)
Price	Cheapest	Expensive	Moderate
Power Use	Uses more power	Uses less power	Uses very little power
Best Use	Storing lots of data	Fast computers and gaming	Portable storage/ removable storage
Examples	Hard disk	Laptop SSD	Pen drive, memory card

1.7 Output Devices

Definition: An output device is a hardware device that presents processed data and information From computer to the user in a human-readable form.

Example: Monitors (LED, LCD), Printer, Speakers, Headphone

Types of Output

1. Soft Copy Output

Output shown on screen or audio

Cannot be touched physically

Examples: Monitor Speaker Projector

2. Hard Copy Output

Output printed on paper

Can be touched physically

Examples: Printer output

Some output devices

1. Monitor

Visual Display Units (VDUs) are output devices used to present processed data to users in the form of text, video, and graphics.

(Monochrome (black-white) and Color monitors)

Type: LED Monitors: Light Emitting Diode

LCD Monitors: Liquid Crystal Display

LCD Monitors	LED Monitors
LCD monitors use liquid crystal technology to display images on the screen	LED monitors use light-emitting diode technology to display images.
Use fluorescent lamps for backlighting; thin, lightweight, energy-efficient.	Use LED lights for backlighting; lighter, thinner, and more energy-efficient than CRT/LCD.

2. Printer

The output device displays data on paper. (Hard copy output)

Types of printers: impact and non-impact printers

Laser Printer

Use laser beam and toner powder to print text and images on paper. Used for Fast, High-quality text documents

Inkjet Printer

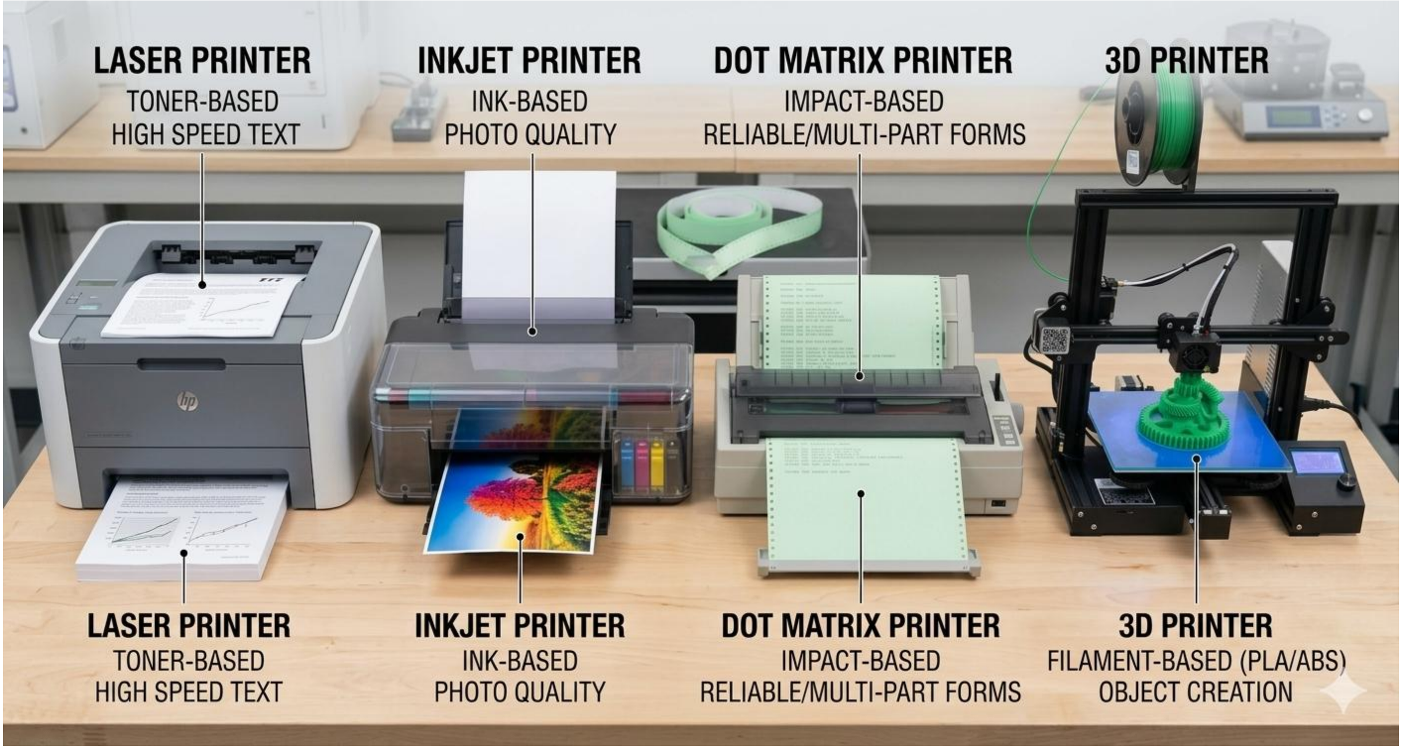
Inkjet printers use liquid ink sprayed to print text and images onto paper.

Use for photos and color printing.

Dot Matrix Printer

Dot matrix printers are impact printers that use pins and an ink ribbon to create dotted characters on paper. Their speed ranges from 50 to 400 CPS. Use for printing bills

3D Printer+: 3D printers are non-impact printers that create three-dimensional objects layer by layer using materials like plastic and metal. Use for making a toy model.



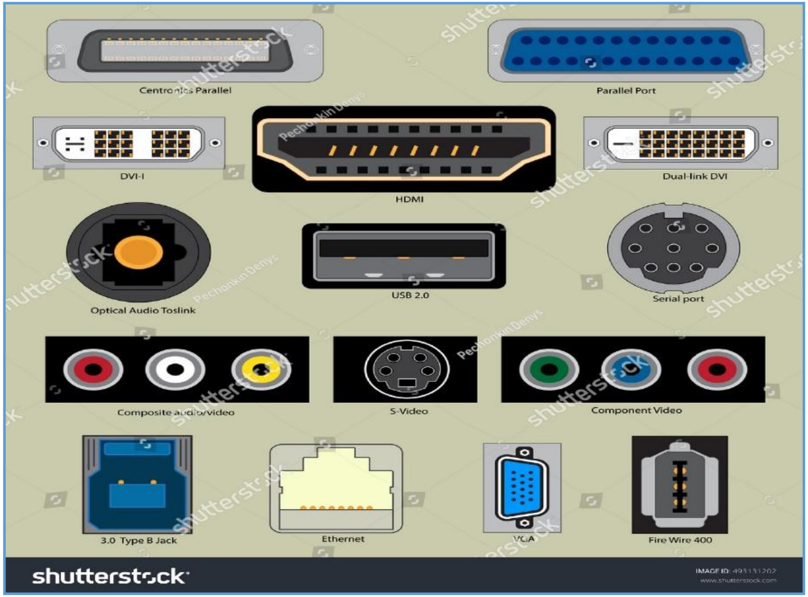
Speaker: Output device that converts digital signals into audible sound.

1.8 Peripheral Devices

Definition: Peripheral devices are external hardware devices connected to a computer to increase its functions and capabilities. For example: Keyboard, Mouse, Printer, Speaker Scanner, Pen drive.

Hardware ports

Hardware ports are connection points used to connect peripheral devices to a computer.



Function of ports

Connects peripherals devices

Supply power

Transfer data

USB Port (Universal Serial Bus)

Used to transfer data over short distances and connect devices like keyboards, mice, printers, and pen drives.

Type C Port (USB Type C)

A small reversible port used for fast charging and high speed data transfer in phones and laptops.

HDMI Port (High Definition Multimedia Interface)

Used to transfer high-quality audio and video through a single cable.

VGA Port (Video Graphics Array)

An older analog port (15 pins) used to connect monitors and projectors in computers.

Ethernet Port

Used to connect a computer to the Internet or local network using a cable

Audio Port

Used to connect headphones, microphones, and speakers to listen or record voice.

Short Question Answer

1. What is RAM? Why is it called volatile memory?

Ans: RAM (volatile) is the primary memory of a computer that stores data and instructions temporarily while processing. It is called volatile memory because data stored in RAM is lost when the power supply is turned off.

2. What is ROM?

Ans: ROM is a type of non-volatile memory, so the stored data remains safe even when the power is turned off. (This type of memory stores BIOS information needed to start the computer). Can read data only.

3. Explain the difference between RAM and ROM in terms of functionality.

Ans: RAM is volatile; ROM is non-volatile.

RAM can read and write; ROM can read only.

RAM stores temporary data; ROM stores permanent BIOS firmware.

In RAM data is lost when power off; in ROM data remains safe when power off.

4. What is the primary function of the CPU?

Ans: The CPU (Central Processing Unit) is the brain of the computer. Its primary functions are: **Fetching** instructions from RAM, **Decoding** them, **Executing** operations (arithmetic and logical), **Storing** results and **managing interrupts**.

5. What is the role of a motherboard in a computer system?

Ans The role of a motherboard in a computer system is to connect all parts of the computer, such as the CPU, RAM, storage devices, and input/output devices, and allow them to communicate with each other.

6. How does a hard drive differ from RAM in terms of data storage?

Ans: A hard drive stores data permanently, while RAM stores data temporarily and loses it when the computer is turned off.

7. Explain the concept of cache memory and its importance in CPU performance.
   OR What is cache memory? Why is it used?

Ans: Cache memory is high-speed memory located between CPU and RAM. It stores frequently accessed data and instructions. It is used to make CPU execution faster.

8. What is a Data Bus?

Ans: A Data Bus is an electronic pathway that transfers digital data between CPU, memory, and other components. It is made of parallel lines.

9. .Define input device. Give examples.

Ans: Input devices are tools used to enter raw data and instructions into a computer system. For example: Keyboard, mouse, microphone, joystick, scanner etc.

10. .What is an output device? Give examples

Ans: An output device is a hardware device that presents processed data and information from computer to the user in a human-readable form. Examples: Monitor (VDU), Printer, Speaker, Projector, Headphone, Plotter

11. What is firmware?

Ans: The BIOS program stored in ROM, which helps to start the computer, is called firmware.

12. What is a peripheral device, and how does it enhance a computer system?

Ans: Peripheral devices are external hardware devices connected to a computer to increase its functions and capabilities. It enhances a computer system by adding extra functions like input, output, or storage.

13. Define Cloud Storage.

Ans: Cloud storage is the online storage of data on remote servers that can be accessed through the Internet. Examples: Google Drive, Dropbox, Microsoft One Drive, Amazon S3

Long Question /Answer

1. Differentiate between primary and secondary memory with examples
2. Describe the roles of cache memory, RAM, and storage devices. How do they contribute to Overall system performance?

Ans:

Cache Memory: Cache memory is a high-speed memory located between the CPU and RAM. It stores frequently used data and instructions so the CPU can access and process them quickly. The CPU first searches data in cache memory; if not found, it searches in RAM and then loads it into cache for faster access.

RAM: RAM is volatile memory that temporarily stores data and instructions during processing. All data in RAM is erased when the computer is turned off. It allows data to be read and written quickly from any location.

Storage Devices: Storage devices store data and instructions permanently. Devices such as HDDs, SSDs, CDs, and DVDs keep data safe even when the power is turned off. Faster storage devices improve loading and startup speed.

"Overall, cache memory speeds up CPU processing, RAM helps run programs smoothly, and storage devices keep data permanently. Together, they improve the overall performance and efficiency of the computer system.

3. Write the difference between CD and DVD.
4. What are the components of the CPU? Explain in detail.
5. Write the difference between SRAM and DRAM.

1.9 Computer Software

Definition: Computer software is a set of instructions, programs, or data that tells computer how to perform specific tasks or functions. It works together with computer hardware to run programs and help users to interact with the computer.

Software is classified into two main types:

- System Software: Helps users to interact with the computer and provides a platform to run other applications, manages hardware resources like memory, i/o device.
- Application Software: Performs specific user tasks

A. System Software:

Definition: System software is the backbone of a computer. It acts as a bridge between the Users and the computer and provides a platform to run other applications.

Types:

(i) Operating system:

It is system software that controls and manages the overall operations of a computer, like a traffic policeman controls traffic. It is the first software loaded into RAM when the computer is turned on.

- It acts as an interface between the user and hardware.
- It makes hardware usable and provides an environment for running software.
- Examples: **Windows, mac OS, Linux, Ubuntu, Android, IOS, UNIX**.

(ii) Language Processor (Translator)

It is system software that translates programs written in assembly language or high-level language into machine language, so the computer can understand them.

Types:

Assembler: Converts assembly language into machine language

Compiler: Converts the whole high-level program into machine language at once

Interpreter: Converts a high-level program into machine language line by line

(iii) Utility Software

It is system software used to maintain and manage computer hardware and data.

Examples: Antivirus, WinZip, WinRAR, Disk Defragmenter, CCleaner, Download Accelerator

(iv) Device Drivers

It is system software that controls and manages a specific hardware device. It makes the hardware usable for the computer.

- Every device (printer, sound card, display card) has its own driver
- Must install driver before using a new device

B. Application Software

Definition: Software designed to perform specific tasks for users, such as writing documents, editing photos, creating presentations, listening to music, or video editing.

Types: **(i) Packaged Software**
(ii) Customized (Tailored) Software

Packaged/general purpose software	Customized/tailored software
Ready-made software designed for general users. Developed by software companies like Google and Microsoft.	Software made for the specific needs of a particular person, organization, or business.
Examples: MS Office, Adobe Photoshop, VLC Media Player, Zoom.	**Examples:** School Management System, Hospital Management System, Mobile Banking.

Open source and proprietary (closed source) software (Based on theaccessibility of source code)

Open Source Software (OSS)	Proprietary Software (CSS)
Source code is publicly available	Source code is NOT available to users
Anyone can view, modify, and use freely	Only the creator/authorized organization can modify
Free or nominal cost	Requires purchasing a license
More flexible and encourages innovation	Limited scope for modification
Less secure (code is open)	More secure (controlled access)
Examples: Linux, Android, Apache, Audacity	**Examples:** Windows OS, Adobe Photoshop, Microsoft Office, AutoCAD, iTunes

Comparison between the features of system and application software

System software	Application software
Manages hardware resources	Fulfills end-user requirements
Written in low-level language	Written in high-level language or 4GL
Runs from power-on to power-off	Runs when the user starts it; stops when the user closes it
Installed during OS installation	Installed as per user's need
Can run independently	Cannot run without system software

Introduction to mobile and web applications

Mobile App

Mobile apps are software designed for smartphones, tablets, and smart watches. They are used with touchscreens, GPS, and cameras.

Examples: Games, social media apps, navigation apps, p

Web Application

Web applications are software that run on web servers and are accessed through a web browser. They do not require installation and can be used on any device with an internet connection.

Examples: Gmail, Google Docs, Facebook, YouTube


Mobile
Application
VS
Web
Application

Short Question Answer

Q1. What is open source software? Give examples.

Ans: Open source software is software whose source code is publicly available, allowing anyone to view, use, modify, and share it freely.

Examples: Linux, Android, LibreOffice, Apache, Audacity

Q2. What is system software? Give examples.

System software is the backbone of a computer. It acts as a bridge between the user and the computer and provides a platform to run other applications.

Examples: Windows, Linux, mac OS,

Q3. What is application software? Give examples.

Application software is software designed to perform specific tasks for users, such as writing documents, editing photos, creating presentations.

Examples: Microsoft Word, Adobe Photoshop, VLC Media Player, Zoom, and Google Meet.

Long Answers Questions.

Q1. Explain the types of computer software with their respective functions.

Ans: Computer software is divided into two main categories.

A. **System Software:** System software is the backbone of a computer. It acts as a bridge between the Users and the computer and provides a platform to run other applications. Types include:

- **Operating system**:

It is system software that controls and manages the overall operations of a computer, like a traffic policeman controls traffic. It is the first software loaded into RAM when the computer is turned on. Examples: Windows, mac OS, Linux,

- **Language Processor (Translator)**

It is system software that translates programs written in assembly language or high-level language into machine language, so the computer can understand them.

Types: Assemble, Compiler, Interpreter:

- **Utility Software**

It is system software used to maintain and manage computer hardware and data.

Examples: Antivirus, WinZip, WinRAR, Disk Defragmenter.

- Device Drivers

It is system software that controls and manages a specific hardware device. It makes the hardware usable for the computer.

B. Application Software: Designed for specific user tasks.

Types include:

- Packaged Software: Ready-made software for general use. Examples: MS-Office, Adobe Photoshop, VLC Media Player, Zoom.
- Customized Software: Developed for specific organizational needs. Examples: Hospital Management System, Banking Software, School Management Software.

Q2. Differentiate between system software and application software based on their roles in a computer system.

Q3. Differentiate between open-source software and proprietary software, giving examples of commonly used applications.

Chapter Summary

Quick Revision

Key Points to Remember

Computer Definition:	Electronic device: Input → Process → Output → Storage
Features:	Automatic, Speed, Accuracy, Storage, Diligence, Versatile
Speed Units:	ms > µs > ns > ps > fs (smallest is fastest)
IPOS:	Input, Process, Output, Storage — basic working cycle
GIGO:	Garbage Out (wrong input = wrong output
Input Devices:	Keyboard, Mouse, Scanner, Microphone, Webcam, Joystick, Touchscreen
CPU Components:	CU (Control Unit), ALU (Arithmetic Logic Unit), Registers
CPU Functions:	Fetch, Decode, Execute, Store, Manage Interrupts
Motherboard:	PCB/backbone; connects all components
RAM:	Volatile; temporary; R/W memory; lost when power off
ROM:	Non-volatile; read-only; stores BIOS; types: PROM, EPROM, EEPROM
Cache:	Fastest memory between CPU and RAM; speeds up processing
Memory Units:	Bit → Nibble → Byte → KB → MB → GB → TB → PB → YB
HDD vs SSD;	SSD is faster, more reliable, no moving parts; HDD is cheaper

CD vs DVD:	CD = 700MB; DVD = up to 17.08 GB
Output Devices:	Monitor (VDU), Printer, Speaker
Soft Copy	Output on screen (monitor)
Hard Copy:	Output on paper (printer)
System Software:	OS, Language Processors, Utility Software, Device Drivers
Application Software:	Packaged (MS-Office) and Customized (Hospital Software)
Open Source:	: Free, modifiable source code — Linux, Android, LibreOffice
Proprietary:	closed source — Windows, MS-Office, Adobe Photoshop

THE END